DYNAMICAL SYSTEMS: CLASS SEPTEMBER 1, 2026
4. NONDIMENSIONALITY AND BISTABILITY.

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Skill Check practice

(1) Plot the function
$$f(N) = H\frac{N^2}{A^2+N^2}.$$
Label your axes and include at least one (labeled) tick mark on each axis.

Your plot should have correct behavior for $N \to 0$, for $N \to \infty$, and for an appropriate (and labeled) finite value of N.

Skill Check practice solution

(1) One way to think of this function is as
$$f(N) = H\frac{(N/A)^2}{1+(N/A)^2},$$
where N/A is dimensionless. Let $x = N/A$.

The N-axis (horizontal) is naturally measured in increments of A. The vertical axis is naturally measured in increments of H.

As $x \to \infty$, we have
$$\lim_{x\to\infty} H\frac{x^2}{1+x^2} = H.$$
At $x = N/A = 0$ we have $f(0) = H\frac{0}{1+0} = 0$. At $x = 1$, so $N = A$, we have
$$f(A) = H\frac{A^2}{2A^2} = \frac{H}{2}.$$

The slope as $N \to 0$ is given by
$$f'(N)\big|_0 = H(2N)(A^2+N^2)^{-1} - HN^2(2N)(A^2+N^2)^{-2}\Big|_{N=0} = 0.$$
Or, using the binomial expansion to approximate the function near 0,
$$f(N) = H(N/A)^2\big(1+(N/A)^2\big)^{-1} \approx H(N/A)^2\big(1-(N/A)^2+\cdots\big)$$
(we have $N/A \ll 1$ when N is close enough to 0). For N close to zero, this is $f(N) \approx HN^2/A^2$, a parabola.

We are now ready to sketch. The curve passes through $(0,0)$ and leaves the origin horizontally (tangent to the N-axis). Close to 0, $f(N) \approx HN^2/A^2$, so it specifically

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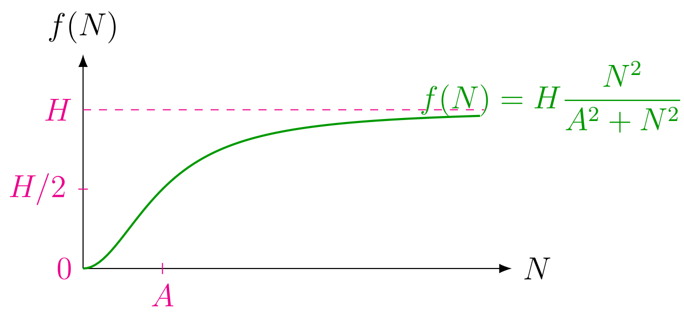
leaves the origin with a parabolic shape. It passes through $f(A) = H/2$ and approaches H as $N \to \infty$.



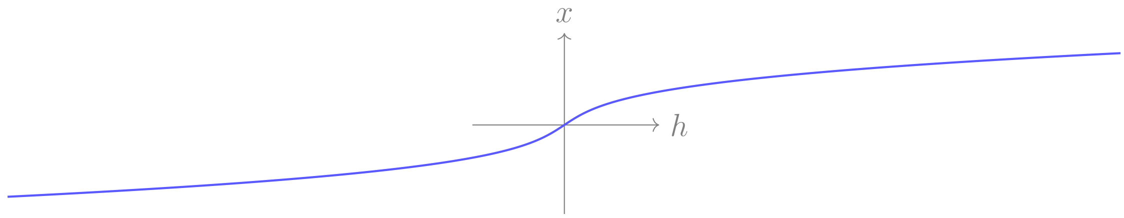
Extra vocabulary / extra facts:

- A **parameter space** is a space where each axis is a parameter.
- A **bifurcation curve** is a curve in parameter space where, at every point along the curve, the associated parameter set is a bifurcation point (meaning that a bifurcation occurs in the system at that set of parameter values).
- A **stability diagram** shows bifurcation curves plotted in a parameter space. The bifurcation curves split the parameter space into regions with qualitatively different phase portraits.
- When a dynamical system, $\dot{x} = f(x)$, has two stable states it is called **bistable**. When it has two or more stable states it may be referred to as having **multiple stable states**.

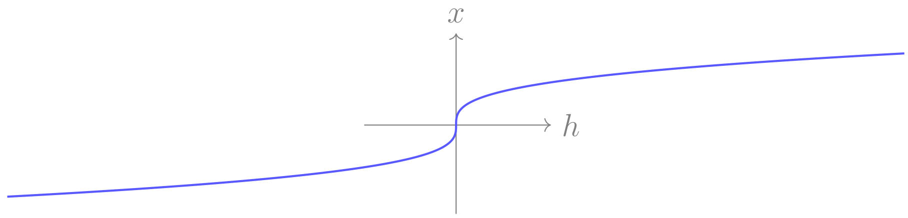
Example with a cusp bifurcation

$$\dot{x} = h + rx - x^3$$

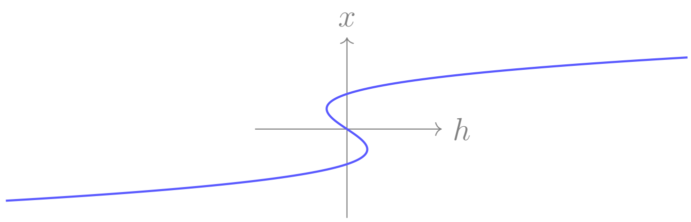
The bifurcation diagrams are made for $r = -1.5, 0, 1.5, 3$. You will see that for $r < 0$ there are no bifurcations in the system. For $r > 0$ there are two saddle-node bifurcations. These saddle-node bifurcations move apart as r increases.



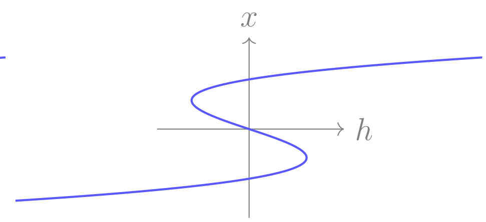
$r = -1.5$



$r = 0$

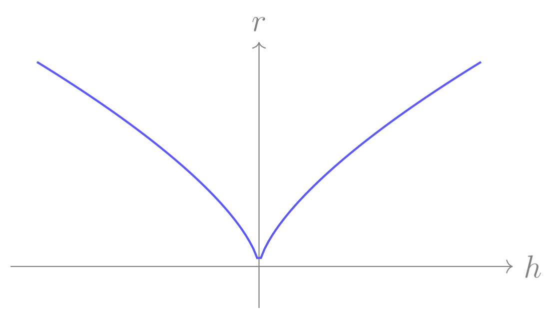


$r = 1.5$



$r = 3$

This stability diagram shows the location of the saddle-node bifurcations in hr-space.



Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- `teamXX-YY.EXT`

where `XX` is your two-digit team number (e.g. `06` if you are team 6), `YY` is the two-digit number of the image for today (e.g. `01` for the first image, `02` for the second, and so on), and `EXT` is the extension for the image filetype you used.

Team Activity

Take your roles from the end of the last class guide and rotate them: questioner → timekeeper → scribe

(1) Consider the nondimensionalized fish population model

$$\dot{x} = x(1-x) - h,$$

where h is a harvesting term.

In this model there is not a feedback between the population of fish (x) and the harvesting term (h). To identify the bifurcation structure of this system, sketch $x(1-x)$ and h both vs. x.

(a) What kind of bifurcation occurs in this system?

(b) Let $h = h_c$ denote the bifurcation point. At $h < h_c$ and $h > h_c$, what is the long-term behavior? (i.e. what is this model predicting for the fishery?)

In the model above, it is possible for the population to become negative because the harvesting rate does not respond to the fish population: "harvesting" continues in the model even when the fish are gone.

Solution. Write $F(x) = x(1-x) - h$. Fixed points satisfy

$$x^2 - x + h = 0, \qquad x_\pm = \frac{1 \pm \sqrt{1-4h}}{2}.$$

Thus the critical harvesting level is

$$h_c = \frac{1}{4}.$$

(a) At $h = h_c$, the two fixed points collide at $x = 1/2$ and disappear. This is a **saddle-node bifurcation**.
The same conclusion follows graphically: $x(1-x)$ is a downward-opening parabola with maximum $1/4$ at $x = 1/2$, whereas h is a horizontal line. The line intersects the parabola twice when $h < 1/4$, is tangent to it when $h = 1/4$, and does not intersect it when $h > 1/4$.

(b) Since

$$F'(x) = 1 - 2x,$$

for $0 \leq h < h_c$ the lower fixed point x_- is unstable and the upper fixed point x_+ is stable. Consequently,

$$x(0) > x_- \implies x(t) \longrightarrow x_+,$$

while an initial population below x_- decreases toward extinction. Mathematically, the model then continues into negative populations because the constant harvesting term never switches off.
At $h = h_c$, the equilibrium $x = 1/2$ is semistable: solutions starting above it approach it, while solutions starting below it decrease away from it. For $h > h_c$, $F(x) < 0$ for every x, so every nonnegative initial population decreases, reaches extinction, and then becomes negative in the unphysical continuation of the model. Thus harvesting above h_c predicts collapse of the fishery, whereas harvesting below h_c permits a stable positive population only if the initial population lies above the threshold x_-.

$\square$

(2) Consider

$$\dot{x} = x(1-x) - h\frac{x}{a+x}.$$

This harvesting term has a feedback: the harvesting rate now depends on x.

(a) Notice that $x = 0$ is a fixed point of the system. Identify its stability as a function of parameters.
(b) Identify how many fixed points the system can have and find the qualitatively different phase portraits that exist at different parameter sets. *To think about other fixed points, one option is to follow the steps below.*
   - Plot $1 - x$ and $h\dfrac{1}{a+x}$. *Do this plotting via your own reasoning, rather than using a computational tool.*
   - Consider how the shape of $h\dfrac{1}{a+x}$ depends on a and on h. Argue that the system could have one, two, or three fixed points.

(c) How did making the harvesting rate depend on the state of the system change the model predictions?

Solution. Assume $a > 0$, $h \geq 0$, and restrict attention to the physically relevant region $x \geq 0$. Factor the vector field as

$$\dot{x} = x\,G(x), \qquad G(x) = 1 - x - \frac{h}{a+x}.$$

(a) The linearization at $x = 0$ is

$$F'(0) = G(0) = 1 - \frac{h}{a}.$$

Therefore $x = 0$ is unstable when $h < a$ and stable when $h > a$. At $h = a$ it is nonhyperbolic, so the linearization alone is inconclusive. Expanding near zero gives

$$F(x) = \left(1 - \frac{h}{a}\right)x + \left(\frac{h}{a^2} - 1\right)x^2 + O(x^3).$$

Hence, at $h = a$, $F(x) \sim (1/a - 1)x^2$. From the physical side $x > 0$, the origin is unstable if $0 < a < 1$, stable if $a > 1$, and, when $a = 1$, one has $F(x) = -x^3/(1+x) < 0$, so it is stable.

(b) Besides $x = 0$, positive fixed points satisfy

$$1 - x = \frac{h}{a+x},$$

or equivalently

$$x^2 - (1-a)x + (h-a) = 0.$$

Thus

$$x_{\pm} = \frac{1 - a \pm \sqrt{(1+a)^2 - 4h}}{2}.$$

The saddle-node value for the two nonzero equilibria is

$$h_{\mathrm{SN}} = \frac{(1+a)^2}{4}.$$

The regime that produces three nonnegative fixed points requires $0 < a < 1$. In that case the phase portraits are as follows (arrows indicate the sign of $\dot{x}$):

parameter range	nonnegative fixed points	phase line
$0 \le h < a$	$0,\ x_+$	$0 \longrightarrow x_+ \longleftarrow$
$h = a$	$0,\ 1-a$	$0 \longrightarrow (1-a) \longleftarrow$
$a < h < h_{\text{SN}}$	$0,\ x_-,\ x_+$	$0 \longleftarrow x_- \longrightarrow x_+ \longleftarrow$
$h = h_{\text{SN}}$	$0,\ (1-a)/2$	$0 \longleftarrow (1-a)/2 \longleftarrow$
$h > h_{\text{SN}}$	0	$0 \longleftarrow$

Thus, for $a < h < h_{\text{SN}}$, both $x = 0$ and $x = x_+$ are stable, while $x = x_-$ is unstable and separates their basins of attraction. This is the **bistable** regime. At $h = h_{\text{SN}}$, the two positive equilibria meet in a saddle-node bifurcation.

For completeness, if $a \ge 1$, the three-fixed-point regime is absent: when $h < a$ there is one positive equilibrium in addition to the origin, and when $h \ge a$ there is no distinct positive equilibrium (apart from the degenerate coincidence at $a = h = 1$).

(c) Because the harvesting term contains a factor of x, harvesting automatically tends to zero as the fish population tends to zero. Consequently, $x = 0$ is an invariant state: a nonnegative solution cannot cross through zero and become negative. The feedback therefore fixes the most obvious unphysical prediction of the constant-harvest model.

It also creates a new qualitative possibility. For $0 < a < 1$ and $a < h < h_{\text{SN}}$, the model is bistable: a population above the threshold x_- recovers to the positive stable equilibrium x_+, while a population below x_- collapses to extinction. Hence the eventual outcome depends not only on the parameters but also on the initial fish population, producing a tipping threshold and hysteresis-like behavior.

$\square$